ALICE L WALTON SCHOOL OF MEDICINE

Cardiac and Vascular Function Curves

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Adapted from: Cardiac and Vascular Function Curves

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Determine the factors that affect cardiac function.
2. Interpret how cardiovascular factors alter vascular function.
3. Apply steady-state operating point mechanisms to describe normal cardiac and vascular function.
4. Predict how changes in contractility, blood volume, venous tone, and afterload affect cardiac and vascular function.

CASE CONNECTION

JH, a 23-year-old male, is evaluated by EMTs after a car crash. He reports severe left-sided abdominal pain from hitting the steering wheel and feels dizzy. His vital signs are: temperature of 98.9°F, low blood pressure (90/60 mm Hg), tachycardia (125 bpm), and normal respiratory rate (16 breaths/min). After the paramedics initiated IV fluids, JH was transported to the hospital. In the emergency department, you suspect a lacerated spleen, causing intra-abdominal bleeding.

What would happen to JH's peripheral vascular resistance if vasopressors are initiated? Consider your answers as you read, and we'll revisit JH at the end of the brick.

GO TO CONCLUSION ↓

What Are Cardiac and Vascular Function Curves (LO1)?

Cardiac physiology requires an understanding of how cardiac output (CO, blood volume pumped per unit of time) changes to meet bodily demands. The CO can be altered by BOTH the blood flowing *into* the heart (venous return) and *out of* the heart (CO) (Figure 1). In this brick, we will learn how the parameters- blood volume, venous tone, contractility, preload, and afterload- interact and change CO. This is precisely what is visualized in the cardiac function and vascular function curves. (For [illegible] tractility, Preload and Afterload brick).

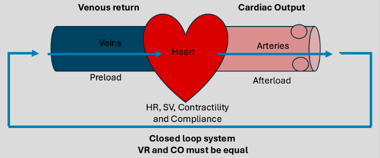


Figure 1. Relationship between venous return and cardiac output, where CO and VR must be the same over time

What is the relationship between cardiac output and venous return?

What Are Cardiac Function Curves (LO2)?

Remember the **Frank-Starling law (of the heart)**? Basically, it indicates that when stretched, cardiac muscle increases the force generated, with the ventricular stretching due to higher blood volumes (venous return) and increased preload (end-diastolic volumes). Mechanically, there is a greater overlap between the actin and myosin filaments in the sarcomeres, more crossbridges, and greater Troponin C sensitivity to calcium. Thus the greater contraction increases the volume ejected with each heartbeat (stroke volume).

This relationship is graphically represented as the **Frank-Starling curve**, with preload (left ventricular end-diastolic volume, LVEDV) on the x-axis and typically stroke volume on the y-axis. However, when heart rate is factored in, cardiac output replaces stroke volume on the y-axis, creating the **cardiac function curve** (Figure 2).

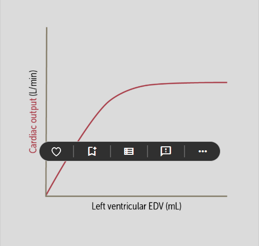


Figure 2 Cardiac function curve

The cardiac function curve serves as a visualization of blood flowing out of the heart into the arteries. As such, it is a visual guide as to how different factors alter the shape of the curve, including cardiac force modulators discussed previously (contractility, preload, and afterload). Additionally, dysfunction or disease can be visualized on the curve, such as the effects of medications and myocardial infarction, hypertension, and heart failure.

Looking at the cardiac function graph, cardiac output is altered by what factor?

How Contractility Affects Cardiac Output– Cardiac Function Curve

Both contractility and afterload affect the cardiac function curve. Let's look at contractility first (Figure 3). With increased preload (or end-diastolic volume EDV), there is higher stroke volume and cardiac output, reflecting the increased contractile force of the heart.

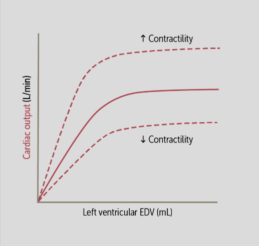


Figure 3. Effect of changed contractility on cardiac function curve (normal curve = solid red line)

CLINICAL CORRELATION

Inotropic drugs such as dobutamine or digoxin, will increase the contractility of the heart. They will thus shift the cardiac function curve upward (the slope steepens), increasing the cardiac output for a given preload.

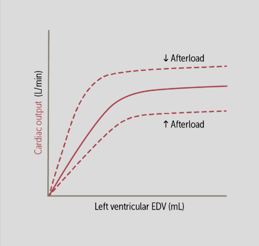
With **heart failure (HF)**, the contractility of the heart is decreased, and the cardiac function curve will move downward (the slope flattens), reducing the cardiac output for a given preload.

How does a decrease in contractility shift the cardiac function?

How Afterload Affects Cardiac Output – Cardiac Function Curve

Afterload is the resis ct against to overcome arterial pressure to eject blood through the semilunar valves into the great vessels. An increase in afterload shifts the curve downward, lowering cardiac output (Figure 4). Decreased afterload shifts the curve upward, improving cardiac output, it is easier to pump blood out into the arteries (Figure 4).

Why is CO decreased with increased afterload? Higher afterload means that the ventricles have to contract more in order to generate these more significant pressures just to open the valves, and there is thus less energy left over to move the blood into the arteries. Thus, this is why changes in afterload produce effects similar to changes in contractility Figure 3 above.



How does a decrease in afterload shift the cardiac function?

What Are Vascular Function Curves (LO3)?

Above, we talked about the amount of blood coming out of the heart (CO), and the ejection of blood into the aorta. That blood eventually returns to the heart via the venous system. So now, let's discuss the amount of blood coming **into** the heart as measured by the venous return. **Venous return (VR)** is the volume of blood returning to the heart per unit time.

A pressure gradient drives venous return. The greater the pressure difference between the **venous system** and **right atrium**, the greater the blood flow back to the heart. Importantly, elevated right atrial pressure reduces this gradient, decreasing venous re[illegible]sed jugular venous pressures in the Cardiac Cycle brick.

This relationship between venous return and right atrial pressure is graphed as the **vascular function curve** (also called venous return curve) (Figure 5). The curve is the same shape whether we use different x-axis parameters: right atrial pressure or end diastolic volume (EDV).

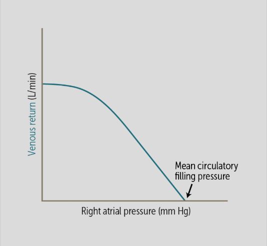


Figure 5. Vascular function curve

So why does right atrial pressure decrease with increased venous return? Right atrial p[illegible]ous return due to a vacuum effect. As the heart pumps more, the increased CO or output creates a stronger "pulling force", which in turn increases venous return (sucking blood in). In fact, veins can collapse if the pressure drops too low (see flat top part of the curve). An analogy is a vacuum machine: one side blows out air (CO), the other side "sucks' air in (VR).

Looking at the vascular function graph, venous return altered by what factor?

Mean Circulatory Filling Pressure (MCFP)

On the vascular function curve, at maximum right atrial pressure (preload) the venous return is at 0 mmHg (where the curve intersects the x-axis) in Figure 5. This specific right atrial pressure is called the **mean circulatory filling pressure (MCFP)**.

Mean Circulatory Filling Pressure (MCFP) is the theoretical equilibrium pressure throughout [illegible]*art stops pumping.* Think of it like water in a sealed container – without the heart creating pressure gradients, blood pressure equalizes across all vessels, and pushes on the walls. Blood volume and venous tone are the primary determinants of MCFP; increasing either raises this baseline pressure. This fundamental concept explains why adequate blood volume and vascular tone are essential for maintaining effective circulation and cardiac output —increasing either raises MCFP.

The vascular function curve is affected by factors such as **blood volume**, **venous tone,** and arteriolar resistance (**afterload**). How these factors affect the **vascular function curve** can readily be visualized as changes in the shape of the curve.

CLINICAL REASONING

Clinical importance: When right atrial pressure increases, the pressure gradient between veins and the heart decreases, making it harder for blood to return to the heart, and blood accumulates in the venous system. In heart failure, the heart's ineffective pumping causes pressure to back up into the venous system (increased jugular venous pressures) and compromises circulation.

How can the MCFP be defined on a vascular function curve?

How Blood Volume and Venous Tone Affect Venous Return

Venous return depends on the pressure gradient between the veins and the right atrium. Factors that increase this pressure gradient, and the resultant venous return, are below and in Figure 6:

- **Volume:** Increase in blood volume (e.g., by infusing normal saline, or venoconstriction).
- **Tone:** Increase in tone of the venous smooth muscle (e.g., venoconstriction from sympathetic activation of the veins)

In conditions with **high blood volume or high venous tone**, the venous return is higher at any given right atrial pressure (preload), and the curve shifts right (Figure 6). The reverse occurs in states of **low blood volume or low venous tone**: the venous return is lower at any given right atrial pressure and the curve shifts left (Figure 6).

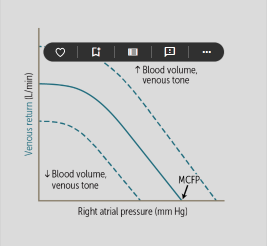


Figure 5. Effect of blood volume and venous tone on vascular function curve

The effect of **venous tone**, or venoconstriction/dilation on venous return are effective because there is a large volume of blood in the large capacitance veins, which can be driven to the heart to increase blood volume rapidly. (The capacitance veins act as a blood reservoir and contain approximately 60% of the total blood volume at rest.)

Notice that the **MCFP** (x-intercept of the curve) also changes in these situations. This happens because an increase in blood volume or venous tone results in a higher venous pressure, and a higher right atrial pressure is required to reduce the venous return to 0 mmHg (Figure 6). The same rationale applies in states of low blood volume or low venous tone, where the MCFP decreases.

How does an increase in blood volume or venous tone shift the vascular function?

How Afterload Affects Venous Return

Now let's see how the resistance of the arterioles (**afterload**) affects the **venous return**. Remember that the **arterial resistance** is determined by the **tone** (degree of constriction) of the smooth muscle within the wall of the arterioles.

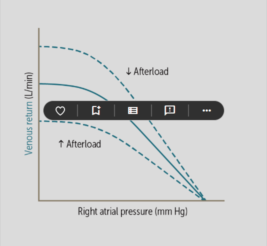


Figure 6. Effect of afterload on vascular function curve

High afterload (increased resistance from arteriolar constriction) restricts blood flow to the venous circulation, causing blood to accumulate in the arterial system. This reduces venous return at any given right atrial pressure, rotating the vascular function curve downward with a flatter slope (Figure 7).

Conversely, **low afterload** (decreased resistance from arteriolar dilation) allows more blood to reach the venous circulation instead of being trapped arterially. This increases venous return at any given right atrial pressure, rotating the curve upward with a steeper slope (Figure 7). Note that total blood volume is not altered by afterload, and **the simply line pivots around the MCFP on the x-axis!**

Why is the MCFP (x-intercept) not affected by changes in afterload? This is because the main determinants of the MCFP are the total blood volume and venous tone, and these factors are not changed by afterload. A simplification that may help with this concept is that one can generally equate **a*R*teries with *r*esistance** and ***V*eins**

How does an increase in afterload shift the vascular function?

What Happens When We Combine the Cardiac and Vascular Function Curves? – Guyton Crossplots (LO4)

Because the **cardiac output** and **venous return** are closely interconnected, the cardiac and vascular function curves can be depicted on the same graph to show their interaction (Figure 8).

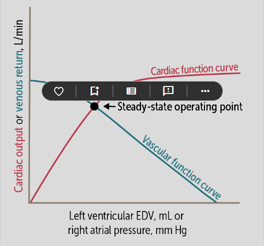


Figure 7 Combined cardiac and vascular function curves

Note that the x-axis can be left ventricular EDV or right atrial pressure but could just as well say “preload.” The y-axis is either cardiac output or venous return. This allows both curves to be plotted on the same graph.

The curves intersect at only one point (black dot on the graph), which represents the equilibrium of the cardiovascular system and is known as the **steady-state operating point**. At this point, the **cardiac output matches the venous return,** and the system operates at a steady state.

Combining the curves allows us to analyze all the major cardiovascular changes on one graph. This is because:

- The cardiac function curve is altered by changes in both contractility and arteriolar resistance (afterload).
- The venous function curve is altered by changes in blood volume, venous tone, and arteriolar resistance (afterload).

Note that afterload changes will modify both curves. More on that in a minute.

What does the intersection between the cardiac and vascular function curves represent?

How Do Contractility, Blood Volume, Venous Tone, and Afterload Affect the Steady–State Operating Point (LO3)?

Now, let’s look at what happens on the combined cardiac and vascular function curves graphs when contractility, blood volume, venous tone, and afterload change. On each graph, the solid red line represents the cardiac function curve, and the solid blue line represents the vascular function curve before we apply any changes. The intersection between the two (black dot) is the initial steady-state operating point.

Changes in Contractility– Crossplot

In Figure 9, we superimpose the cardiac function curves on the vascular function curve to see how a change in contractility affects the steady-state operating point.

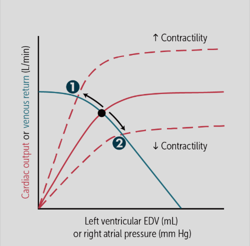


Figure 9. Effect of contractility on steady-state operating point

Increased Contractility. With higher contractility, the cardiac function curve steepens (upper dotted red curve). Since contractility does not affect the vascular function curve, the blue curve will remain the same. The result is **point 1**, the new steady-state operating point, where the cardiovascular system reaches a new steady- [illegible] eturn) at a lower preload. At point 1, both the cardiac output and the venous return are higher.

Decreased Contractility. What if myocardial contractility falls (eg, after a heart attack)? The reverse happens, and the cardiac function flattens (lower dotted red curve). This brings us to steady-state operating **point 2**, with the equilibrium being reached at a higher preload. At point 2, both the cardiac output and the venous return are lower.

CLINICAL REASONING

The crossplots can be used to evaluate heart failure. In heart failure, cardiac output decreases, right atrial pressure rises, and venous return decreases. These changes result in the backing up of blood in the venous system, and patients develop jugular venous distention and peripheral edema.

Changes in Blood Volume and Venous Tone– Crossplot

In Figure 10, we see how changes in blood volume and venous tone affect the steady-state operating point.

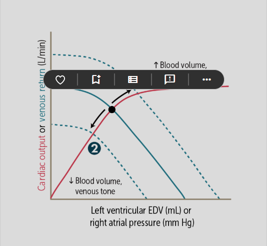


Figure 9 Effect of blood volume and venous tone on steady-state operating point

Increased Blood Volume or Venous Tone. Let's say a patient gets a blood transfusion, which increases blood volume. This shifts the vascular function curve up and to the right (upper dotted blue curve) while the cardiac function curve remains unchanged. A new steady state (CO = venous return) is established at a higher preload, depicted by **point 1**. At this new steady-state operating point, at a higher preload, both cardiac output and venous return increase. The **MCFP** (x-intercept) also increases. An isolated increase in venous tone would have the same effect.

Decreased Blood Volume or Venous Tone. Conversely, blood loss has the opposite effect. It shifts the curve down and left (lower dotted blue curve). A new equilibrium is established at a lower preload, depicted by **point 2.** The result would be lower cardiac output and venous return at a lower preload and decreased **MCFP.** An isolated decrease in venous tone would have the same effect.

Once caveat, blood loss would trigger additional compensatory changes to increase blood pressure, including HR and contractility, as well as arterioconstriction (afterload).

CLINICAL CORRELATION

Blood volume can also change with isolated changes in the volume of plasma. This might occur upon saline infusion (increases the blood volume) or diuretic use (decreases the blood volume).

Changes in Afterload – affect BOTH cardiac and vascular function curves

The effect of changed afterload on the combined graph is more complex than effects of the other factors, because changes in afterload affect **both** the cardiac and the vascular function curves at the same time (Figure 11). Notice that, as indicated previously, changes in afterload **do not alter the MCFP, where the vascular function curve intersects the x-axis.**

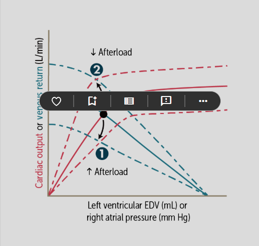


Figure 10 Effect of afterload on steady-state operating point

Increased Afterload. Let's hypothesize that the patient takes a vasoconstricting drug, that constricts the arterioles and increases the afterload. Both the cardiac output and the vascular function curves will rotate lower (lower dotted curves), and a new steady-state operating point (**point 1**) will be established in which both the cardiac output and the venous return are lower. The end-diastolic volume (EDV) has not changed much.

Decreased Afterload. On the other hand, if the patient takes a vasodilator drug, such as a calcium channel blocker, the arterioles will dilate, and afterload will decrease. In this scenario, both the cardiac and the vascular function curves will rotate upwards (upper dotted curves), and a new steady-state operating point (**point 2**) will be established in which both the cardiac output and the venous return are higher, with minimal change to EDV.

CASE CONNECTIO

K TO INTRODUCTION ↑

Thinking back to JH. Because JH has uncontrolled internal bleeding, the blood volume in his veins decreases, leading to decreased venous return, preload, cardiac output, and low blood pressure. If vasopressors are initiated, peripheral resistance and blood pressure would increase. However, JH also needs to have the bleeding quickly controlled to correct his altered hemodynamic state. You simultaneously order an abdominal CT and consult trauma surgery. JH underwent a splenectomy, recovered without any complications, and was discharged on hospital day 4.

Summary

- The cardiac function curve is a graphical representation of the relationship between cardiac output (CO) and preload (represented by left ventricular end-diastolic volume).
- An increase in contractility or a decrease in afterload steepens the cardiac function curve, whereas a decrease in contractility or an increase in afterload flattens it.
- The vascular function curve is a graphical representation of the relationship between venous return and preload (represented by right atrial pressure).
- The mean circulatory filling pressure is the right atrial pressure at which venous return becomes 0.
- The steady-state operating point is the intersection between the cardiac output and vascular function curves and represents the cardiovascular

equilibrium at which the CO is equal to the venous return.

- The cardiac function curve is altered by changes in both contractility and arteriolar resistance (afterload).
- The venous function curve is altered by changes in blood volume, venous tone, and arteriolar resistance (afterload).
- Afterload changes will modify both curves.
- An increase in contractility will shift the steady-state operating point up and to the left, reflecting a higher cardiac output and higher venous return at a lower end-diastolic volume (EDV).
- An increase in blood volume will shift the steady-state operating point up and to the right, reflecting a higher CO and higher venous return at a higher EDV.
- An increase in afterload will shift the steady-state operating point down, reflecting lower CO and lower venous return.

Review Questions

1. A 67-year-old male with an inferior wall myocardial infarction has proximal right coronary artery occlusion. What is the effect on the heart contractility and cardiac output?

A. ● Decreased contractility, decreased cardiac output

B. ○ Decreased contractility, increased cardiac output

C. ○ Increased contractility, decreased cardiac output

D. ○ Increased contractility, no change to cardiac output

E. ○ No change to contractility, no change to cardiac output

Hide Explanation

Correct answer A). Because of the relationship between right ventricular contractility and the volume that ends up in the left ventricle (as well as the decrease in blood pressure), the cardiac output is decreased because of the decrease in stroke volume. Incorrect answers: B). The CO would not increase. C). Contractility describes the contractile force of cardiac muscle, which would be decreased in a patient with myocardial infarction, not increased. D). and E). contractility describes the contractile force of cardiac muscle, so increased contractility, or no change to cardiac output are both incorrect.

2. A 5-year-old girl ingested amlodipine, a calcium-channel blocker that preferentially reduces arteriolar tone. How would this affect her cardiac output and venous return?

A. ○ Decreased cardiac output, decreased venous return

B. ○ Decreased cardiac output, increased venous return

C. ○ Increased cardiac output, decreased venous return

D. ● Increased cardiac output, increased venous return

E. ○ No change in cardiac output, no change in venous return

Hide Explanation

Correct answer D). The vasodilator drug causes arteriolar relaxation, decreasing the afterload. This would increase the volume of blood ejected on each heartbeat (stroke volume), and thus increase the cardiac output. Incorrect answers: A) and B). As indicated above, the CO would increase, not decrease. C). The decrease in arteriolar tone would allow more blood to move from the arterial to the venous circulation, increasing and not decreasing the venous return. E). At the new steady-state operating point, cardiac output and venous return are higher than their initial value.

3. Which of the following events would decrease the mean circulatory filling pressure?

A. ○ Blood transfusion

B. ● Diuretic use

C. ○ Excessive hydration

D. ○ Myocardial infarction

E. ○ Vasopressor use

Hide Explanation

The correct answer is diuretic use B). The mean circulatory filling pressure (MCFP) is the pressure that wo[illegible]ular system if the heart stopped pumping and blood ceased to flow, and its two main determinants are the blood volume and venous tone. A). Blood transfusion and C). excessive hydration would increase the blood volume and therefore the MCFP. D). Myocardial infarction would decrease the contractility of the heart and not affect the MCFP. E). Vasopressor use can constrict both the arteries and the veins. However, an increase in venous tone has the same effect as an increase in blood volume, so the MCFP would increase in this case.

4. An 11-month-old with sickle cell disease presents with lethargy, pallor, and distended abdomen. Vitals show hypotension (70/40 mmHg), severe anemia (hemoglobin 6 mg/dL), and splenomegaly. Diagnosed with splenic sequestration crisis, the infant receives IV saline and blood transfusion. Which statement correctly compares the infant's initial condition on arrival versus the post-intervention?

A. ○ Cardiac output is increased

B. ○ Steady-state operating point occurs at a higher preload

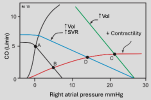
C. ● Steady-state operating point occurs at a lower preload

D. ○ Venous return is increased

Hide Explanation

Correct answer C). Th[illegible]es splenic sequestration crisis, w[illegible]trapped in the spleen, decreasing the blood volume that is available in circulation. Administering fluids would bring the blood volume back closer to normal, allowing steady-state operating point to occur at a higher preload than her initial, lower state. Incorrect answers: A). At the moment of sequestration crisis, CO would decrease, not increase. B). The steady state is caused by a decreased preload, not a higher preload. D). The change is caused by a lower venous return, not increased venous return.

5. A 65-year-old male presents with shortness of breath, leg swelling, and fatigue. On exam, he has jugular venous distention, pitting edema, low blood pressure (90/60 mmHg), and tachycardia (110 bpm). He has severe heart failure, with depressed cardiac function and reduced cardiac output (CO). The body compensates by increasing preload, afterload (resistance), and venous return, but these changes are maladaptive since the heart is already dilated with reduced contractility. In the image below, there are 3 venous function curves (black, blue, and green) and 2 cardiac function curves (black, red). Select the letter that corresponds to the steady-state operating point of this patient.



Credit: Klabunde

A. ○ A

B. ○ B

C. ○ C

D. ● D

Hide Explanation

Correct answer: D) Severe Heart Failure: The steady state point is abnormal. Compensatory mechanisms attempt to restore cardiac output through increased afterload (vascular resistance) and preload (blood volume- increased preload increases contractility- Frank Starling Law of the Heart). However, excessive vasoconstriction and fluid retention (blue line) provide minimal cardiac output improvement. Indeed, the increased volume and pressures worsen the condition, causing edema and increased afterload and work of the heart. Incorrect answers: A). This is the normal steady state point, with normal cardiac output (CO) and low right atrial pressure (RAP) (both black lines). B). The onset of heart failure (HF) is characterized by decreased cardiac function (red line), which leads to a large decrease in cardiac output (CO) and an increase in right atrial pressure (blood backing up due to poor contractility). However, the increased C). Compensated HR - Increased blood volume and venoconstriction (green curve) improve CO. The CO shifts to the right on the contractility curve.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Determine the factors that affect cardiac function.~~

02 ☑ ~~Interpret how cardiovascular factors alter vascular function.~~

03 ☑ ~~Apply steady-state operating point mechanisms to describe normal cardiac and vascular function.~~

04 ☑ ~~Predict~~ ~~, venous tone, and afte~~

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Objective Confidence: 4/4